

(IC2), a powerful magnet and a few discrete components. The sensor is a three-terminal device. The voltage generated by it depends on the polarity of the magnet. The presence of south-polarity magnetic field, perpendicular to the branded face of the sensor, in-

trolled by the sensor output.

When the door is closed, the magnet is close to the sensor. Presence of north-polarity magnetic field generates a low voltage at the output of the sensor. As a result, pin 4 of NE555 also goes low and the timer does not

creases the output voltage of the sensor. Conversely, the presence of north-polarity magnetic field in the same orientation as the branded face of the sensor device decreases the sensor output.

NE555 is wired as an astable multivibrator whose reset pin 4 is con-

oscillate. This de-activates the speaker indicating that the door is closed.

When the door is opened, the magnet moves away from the sensor. The output of the sensor goes high and pin 4 of IC2 also goes high. The timer oscillates and the speaker sounds to indicate that the door is open. The speaker sounds until the door is closed.

Assemble the circuit on a general-purpose PCB and enclose in a suitable case. Fix the Hall sensor at the corner of door frame and the magnet on the door, keeping its north pole (N) oriented towards the sensor. Align the sensor and magnet such that when the door is closed the output of the Hall sensor is low.

The circuit works off regulated 5V DC. An IC 7805-regulated supply can be used for the purpose. ●

PORTABLE SIGNAL WAND

■ T.K. HAREENDRAN

Flashing lights have varied applications. For example, travellers can use these as warning beacons on highways to catch the attention of the public in case of an emergency. Described here is the circuit of a battery-operated, LED-based portable signal wand.

The signal wand is in fact a three-channel, sequential dual-colour LED flasher built around two popular and inexpensive chips—CD4093 and CD4017. CD4093 (IC1) consists of four inbuilt Schmitt-trigger circuits. Each circuit functions as a two-input NAND gate with Schmitt-trigger action on both the inputs. Here one gate (N1) of IC1 is wired as a gated astable multi-

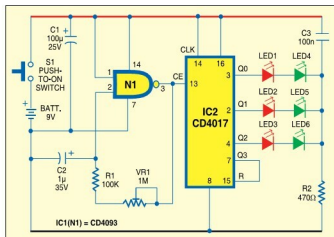


Fig. 1: Circuit for portable signal wand

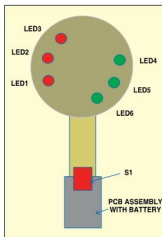


Fig. 2: Assembly of signal wand

vibrator with its control pin 1 permanently connected to the positive supply as shown in Fig. 1. The frequency of the astable multivibrator can be varied with the help of a 1-mega-ohm preset (VR1). CD4017 (IC2) is a five-stage divide-by-ten Johnson counter with ten decoded outputs and a carry bit.

Counter IC2 is cleared to zero count by a logic high on its reset pin. IC2 advances by one count on the positive edge of the clock signal when the clock-enable signal is in the logic-low state. Ten decoded outputs of IC2 (Q0 through Q9) are normally low and reach high state only at their respective time-slot. Each decoded output remains high for a full clock cycle.

The clock output from gated astable N1 is fed to clock-enable pin (pin 13) of IC2. The clock input (pin